READING PROCESS WORKSHEET

GED0001

NAME: _ _ _GROUP 4- Clever Cookies_____ SECTION: __TA01_____

TEXT TITLE:__ Big Bang: Dutch Firm eyes space baby_____ __ DATE: _02/06/2026_ _TEXT TYPE: Scientific Research Paper____

AUTHOR: AFP (Agence France-Presse)

I. PRE-READING

A. Complete the table with the headings, subheadings, and visuals that can be seen in the text.

Headings	Subheadings	Visuals
- Big Bang: Dutch firm eyes space baby	- Entrepreneur opens space sex research to make human interplanetary species - Creating perfect environment - Crazy Ambitions	- A photograph of the space/planetary horizon - The photo describing the Spaceborn United's research on human reproduction in space - AFP Photo credit

B. List down unfamiliar words and words that signal transition (e.g., in addition, however, etc.)

Unfamiliar Words	Signal Words
1. Waning Immunity 2. Downtick Point 3. Compartmental Model 4. Epidemiological Model 5. Omicron Surge	1. However 2. Moreover 3. Therefore 4. Lastly 5. In addition

C. Fill the table with information that you already know about the topic (K), you wonder about the topic (W), and you will learn about the text (L). Do not fill the L column yet.

K	W	L
● Space is not made for humans with things such as gravity, and lack of resources. ● Pregnancy requires heavy attention and care. ● Human reproduction experiments is very controversial.	● Can a baby develop properly in space? ● How will the mother get support while developing the baby? ● What are the ethical issues that could happen?	● It is difficult for sexual intercourse to occur in space due to issues such as the couples drifting away. ● Research is already being done in partial gravity. ● Developing embryos in space can be dangerous because of the hazards in space such as higher radiation than on earth, to gravity that embryos aren't designed for.

II. READING:

A. Annotate (write marginal notes) the text's copy. Use any of the following phrases as prompts in writing your thoughts about the ideas/information found in the text.

1. Make inferences
    - I think the author means...
    - Perhaps the author is suggesting that....
    - This event probably occurred before/during/after... (e.g., World War II)
    - What must have happened could be that...
    - I think it happened because...
2. Question.
    - How...?
    - Why...?
    - What if....?
    - Who...?
    - What could be an example of this concept...?
3. Visualize.
    - I imagine...
    - I can almost/hear/taste/smell/feel...
    - It must look like a…
4. Make connections.
    - This is like...because...
    - This reminds me of... because...
    - This is similar to... because...
    - I read an article/book about this that states...
    - This happened to me when…
5. Determine the importance/relationship of ideas/information.
    - This must be the author's thesis...because....
    - This example supports the author's point about...
    - This seems important because...
    - This figure is used to prove what the author says about...
    - This testimony tries to support the author's claim that…

'Big Bang': Dutch firm eyes space baby

Entrepreneur opens space sex research to make human interplanetary species

EINDHOVEN, NETHERLANDS—Climate crisis, nuclear Armageddon, or a sudden meteor strike—it's clear humanity could do with Planet B. But first we need to learn to reproduce safely in space, says Dutch entrepreneur Egbert Edelbroek.

Edelbroek's firm, Spaceborn United, is pioneering space sex research, with the eventual aim of natural conception and birth in the partial gravity environment found on Mars.

The challenges of achieving safe space sex are galactic, but the ambitious Dutchman is confident he will see an extraterrestial human child born within his lifetime.

"It's important that the Earth and humanity can become a multiplanetary species," Edelbroek told AFP.

"If you want to have independent human settlements beyond Earth, and if you really want them to be independent, you also need to address the reproductive challenges," said the entrepreneur.

Actual sexual intercourse in space presents many difficulties, chief among them the lack of gravity—a couple would drift away from each other—so Spaceborn United is first trying to conceive an embryo in space.

Starting with mice, before eventually moving to human sperm and egg cells, the firm has created a disc that mixes the cells together, with the aim of producing a viable embryo.

It's like a "space station for your cells," said Aqeel Shamsu, CEO of the UK-based Frontier Space Technologies, which is working with Spaceborn on the project.

PLANET B. Dutch firm Spaceborn United is launching research aimed at helping humans reproduce in the weightless environment of outer space. —AFP

This embryo is then cryogenically frozen, to pause their development, but also to protect them during reentry—"It's a lot of shaking, a lot of vibration, a lot of G-forces. You don't want to expose embryos to this," said Edelbroek.

Creating perfect environment

Research is currently under way in simulated partial gravity laboratory conditions but Edelbroek said a launch with mice cells was planned for the end of next year, with a timeline of "about five or six years" for the first launch with a human embryo.

But that's only one small step. A giant ethical leap remains before such an embryo could be implanted back into an Earthling woman to give birth to the first child conceived in space.

"It's a delicate topic. You're exposing vulnerable human cells, human embryos, eventually, to the hazards of space, to radiation that is much higher than on Earth, to different gravity environments that embryos are never designed for," said Edelbroek.

Such ethical issues are one reason why research into space reproduction has generally been left to private firms like Spaceborn, rather than Nasa, which is queasy about spending tax dollars on such sensitive topics.

Edelbroek said his firm was the only one looking to develop a human embryo in space.

Bodily fluids that are pulled down on Earth would be drawn upwards in a low-gravity environment, posing several challenges for the human body.

"An adult body can handle some differences, but you don't want to expose a growing, more vulnerable fetus to these different variables. So you need to create the perfect environment first," he said.

'Crazy ambitious'

One new factor in space reproduction is the growth of space tourism, fueled by companies like SpaceX and Virgin Galactic.

Couples on a space tourism flight might want to go down in history as the first to conceive, warned Edelbroek, adding that he was consulting with the sector to make them aware of the risks.

Spaceborn's research—which replicates the IVF process but in space—is also helping people closer to home to conceive, said Edelbroek.

The Dutchman said he had been forced to scale back his plans—"we've gone from crazy ambitious to just very ambitious"—as the scale of the challenges became clear.

Nonetheless, he is sure that a baby will be born in space within his lifetime.

"I expect to be at least 100 years old," the 48-year-old said. "So that should give us enough decades to achieve that, absolutely."

"Eventually, humanity hopefully will be needs achieve childbirth in space." —A

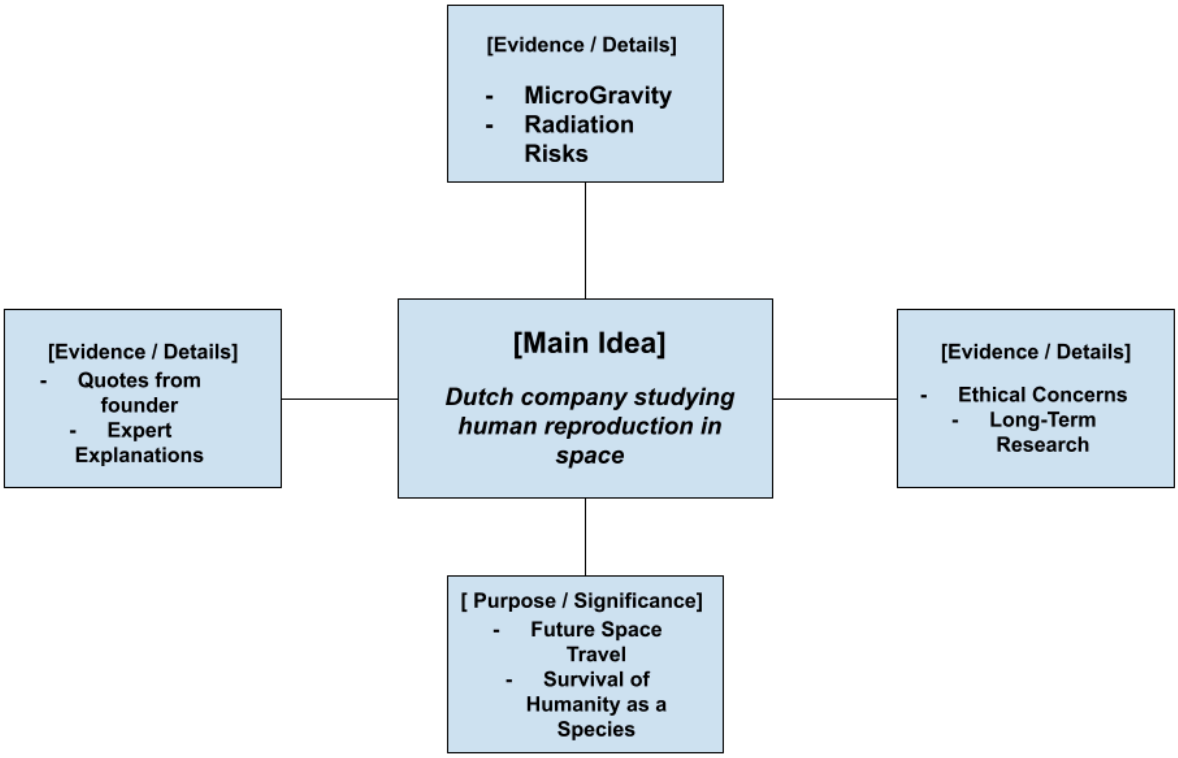
B. Fill the table with specific details from the text and inferences that can be drawn from it.

Details from the Text	Plausible Inferences
- "This embryo is then cryogenically frozen, to pause its development, but also to protect it during re-entry, it's a lot of shaking, a lot of vibration, a lot of G-forces. You don't want to expose embryos to this" (G-Force) - "so Spaceborn United is first trying to conceive an embryo in space." - "the ambitious Dutchman is confident we can see an extraterrestrial human child born within his lifetime."	- Human reproduction in space is explanable scientifically but extremely complex. - G-Force is a constant critical barrier - There is an advancement in technology when it comes to controlled IVF in space

C. Figure out the meaning of the unfamiliar words listed in the Pre-reading part of this worksheet. Write the unfamiliar word, look for the clue, and write your own definition.

Unfamiliar Word	Context Clue (i.e., antonym, synonym, examples, description, word parts, definition or any clues in the text	Your own definition of the word based on the clues
Interplanetary	Make human interplanetary species	Make humans have freedom traveling the Solar System.
Armageddon	Sudden Meteor Strike, Climate crises	End of the world.
Queasy	Queasy about spending tax dollars on such sensitive topics.	Feeling uneasy.
Extraterrestrial	Extraterrestrial human child born within his lifetime	Anything that originates outside of Earth.
Multiplanetary	Earth and humanity can become a multiplanetary species	Building civilization outside Earth.

D. Draw an appropriate graphic organizer below. Use it to organize ideas/information from the text. Use the space on the back of this paper.


[Evidence / Details]
- MicroGravity
- Radiation Risks
[Evidence / Details]
- Quotes from founder
- Expert Explanations
[Main Idea]
Dutch company studying human reproduction in space
[Evidence / Details]
- Ethical Concerns
- Long-Term Research
[Purpose / Significance]
- Future Space Travel
- Survival of Humanity as a Species

III. POST-READING

A. Answer the questions written in the W column in the KWL chart.

Anticipation Questions	Answer is right there in the text	Answer is implied in the text	Answer is not in the text at all
● Can a baby develop properly in space?		The article doesn't give a simple "yes" or "no" because it hasn't been done yet. However, it implies it will be very hard, noting that embryos aren't designed for high radiation and the human body struggles in low gravity.	
● How will the mother get support while developing the baby?			The text only talks about the science and tech side. It doesn't mention how the mother will be cared for or supported.
● What are the ethical issues that could happen?	The text says it's a "delicate topic" and a big risk to put embryos in danger. It also questions if private companies should do this instead of NASA.		

B. Fill the L column in the KWL chart with the ideas/information you have learned after reading the text.

The article explains that a Dutch firm is researching how humans can reproduce in space to help us live on other planets. They are testing ways to mix cells and grow embryos in special labs that mimic space conditions. However, high radiation and low gravity make this very dangerous and difficult for a developing baby. This creates big ethical concerns about the safety of human cells and the role of private companies in such risky research.

C. Answer the following rhetorical analysis guide questions.

1. What credentials does the author have which give him/her the authority to write about the topic of the text?
   - **This article is written not just by one person but rather by an organization. The AFP, or Agence France-Presse, from what I researched, is considered one of the oldest and most reputable news agencies in the world. As a news agency, they hire skilled and experienced journalists whose work meets the industry's standards, and as journalists of such an agency, they have access and connections to experts in various fields, be it from science, education, business, etc. And their works go through rigorous editorial checks, which reduces bias and eliminates misinformation.**

2. After considering the author's profession and affiliation, what possible biases the author might have about the topic?
   - **The bias that may be possible considering the fact that AFP is a news agency is that they have institutional bias. Considering their role in society, they may frame stories about discoveries and progress, just like the article "Big Bang: Dutch Firm Eyes Space Baby," which could exaggerate the progress we are making, but in reality, it's all in the speculation phase.**

3. What is the purpose of the text, and how does the author accomplish that purpose? What evidence does the author use to support the main idea in the text?

- **The purpose of the text is to tell the readers about research about human reproduction in space that a Dutch company is holding. The AFP accomplishes this by showing the problem, describing the research, and adding the opinions of experts and the limits of the study. The main idea is supported by the AFP through expert opinion, detailed description of the planned research, and placing the idea in the context of reality.**

4. What specific idea/information in the text challenges or surprises you? Why?
- **The main idea of having human reproduction in space. It's just that it never occurred to me that it would be a problem and that it's such an outlandish idea that it never really crossed my mind in the first place.**

3. . Is the style of writing suitable for the intended audience? Is it too formal or too casual? Why?
    - **Yes, it's not too formal but not too casual either. Probably because their audience is all around the world, so it would make sense why they would use simple English so that all their readers can understand.**